Prokaryotic Promoters

Framework	Year	Tool	Web-server	Github	Embedding	Algorithm	Evaluation Strategy	Promoter Type	Species	DataBase	Seq. Length
Machine Learning	2016	BacPP neural-network tool	Link	–	One-Hot Encoding	ANN	10-fold CV	σ^{70}, σ^{24}, σ^{32}, σ^{54}, σ^{28}, σ^{38}	E. Coli	–	81 bp
Machine Learning	2017	iPro70-PseZNC	Link	–	PseZNC	SVM	5-fold CV	σ^{70}	E. Coli	RegulonDB	81 bp
Machine Learning	2017	bTSSfinder	Link	–	PWMs	NN Classifier	10-fold Random Testing	σ^{70}, σ^{24}, σ^{32}, σ^{54}, σ^{28}, σ^{38}, σ^{A}, σ^{C}, σ^{H}, σ^{G}, σ^{F}	E. Coli & cyanobacteria	RegulonDB and CollecTF	251 bp
Machine Learning	2018	ipromoter-2L	Link	–	PseKNC	SVM	5-fold CV	σ^{70}, σ^{24}, σ^{32}, σ^{54}, σ^{28}, σ^{38}	E. Coli k-12	RegulonDB	81 bp
Machine Learning	2018	IBPP	Not Available	Link	Image template	SVM with RBF Kernel	–	σ^{70}	E. coli K-12 MG1655	RegulonDB	81 bp
Machine Learning	2018	70ProPred	Link	–	PSTNP + PseEIIP	SVM	5-fold CV	σ^{70}	E. Coli K-12	RegulonDB	81 bp
Machine Learning	2019	iPro70-FMWin	Link	–	k-mer	SVM	5-fold CV	σ^{70}	E. Coli K-12	RegulonDB	81 bp
Machine Learning	2019	iPSW(2L)-PseKNC	Link	–	PseKNC + Physiochemical property	SVM	Cross Valid. with n-fold	Promoter + Strength	E. Coli	RegulonDB	81 bp
Deep Learning	2019	Deep Promoter	Not Available	Link	FastText N-grams	1D CNN	5-fold CV	Promoter + Strength	E. Coli	RegulonDB	81 bp
Machine Learning	2019	MULTiPly	Link	–	KNC + KNN + DAC + BPB	SVM	5-fold CV	Promoter + Strength	E. Coli	RegulonDB	81 bp
Machine Learning	2019	iPromoter-2L2.0	Link	–	k-mer + PseKNC	SVM	5-fold CV	σ^{24}, σ^{28}, σ^{32}, σ^{38}, σ^{54}, σ^{70}	E. Coli	RegulonDB	81 bp
Machine Learning	2019	Promote-Predictor	Not Available	Link	Motif-profiles + derived seq. features	SVM	Cross Valid. with n-fold	σ^{54}	E. Coli	RegulonDB	81 bp
Machine Learning	2019	iPromoter-FSEn	Link	–	feature-subspace based seq. features	Ensemble classifier	5-fold CV	σ^{70}	E. Coli	RegulonDB	81 bp
Machine Learning	2019	Phage-Promoter	Link	Link	Free energy + Motif + Position specific scoring matrix	ANN & SVM	5-fold CV	Promoter or Not	bacteriophages infecting bacteria	phiSITE	25 bp
Deep Learning	2020	iPromoter-BnCNN	Link	–	K-mer nucleotide & structural properties	1D CNN	5-fold CV	σ^{24}, σ^{28}, σ^{32}, σ^{38}, σ^{54}, σ^{70}	E. Coli	RegulonDB	81 bp
Machine + Deep Learning	2020	iPSW(PseDNC-DL)	Link	–	PseDNC	CNN	Cross Valid. with n-fold	Promoter + Strength	E. Coli	RegulonDB	81 bp
Machine Learning	2020	SEProm	Link	–	28 structural & 3 energetic parameters	Statistical-Learning Classifier	–	Promoter or Not	multiple bacterial + archaeal species	Compre-hensive dataset across multiple species	975 nt for the 31-param. vector
Deep Learning	2020	pcPromoter-CNN	Link	Link	One-hot encoding	1D CNN	5-fold CV	σ^{70}, σ^{54}, σ^{38}, σ^{32}, σ^{28}, σ^{24}	E. Coli	RegulonDB	81 bp

Framework	Year	Tool	Web-server	Github	Embedding	Algorithm	Evaluation Strategy	Promoter Type	Species	DataBase	Seq. Length
Machine Learning	2021	Promotech	Not Available	Link	One-hot & Label Encoding	RF + RNN	5-fold CV	Promoter or not	Multiple Species	–	40 bp upstream of TSS
Machine Learning	2021	iPro2L-PSTKNC	Not Available	Link	PSTKNC	SVM classifier	5-fold CV	σ^{70}, σ^{54}, σ^{38}, σ^{32}, σ^{28}, σ^{24}	E. Coli	RegulonDB	81 bp
Machine Learning	2021	iPromoter-ET	Not Available	Link	One-hot + PseKNC encoding	SVM	5-fold CV	Promoter and Strength	E. Coli	RegulonDB	81 bp
Machine Learning	2022	PPred-PCKSM	Not Available	–	PCKSM	ANN	5-fold CV	σ^{70}, σ^{24}, σ^{28}, σ^{32}, σ^{38}, σ^{54}	E.Coli	RegulonDB	81 bp
Machine Learning	2022	PredPromoter-MF(2L)	Not Available	Link	sequence-based features	Ensemble Model	5-fold CV	Promoter or Not	E. Coli	RegulonDB	81 bp
Deep Learning	2022	Expositor	Not Available	Link	Various DNA Encoding	NN	–	Promoter or Not	Multiple Species	–	58 bp
Machine Learning	2022	dPromoter-XGBoost	Not Available	–	k-mer, PseKNC, PseDNC	XGBoost	Cross Valid. with n-fold	Promoter and Strength	Multiple Species	RegulonDB	81 bp
Machine Learning	2022	BERT-Promoter	Not Available	Link	Pre-trained BERT	Machine Learning Classifier	–	Promoter and Strength	E. Coli	RegulonDB	81 bp
Deep Learning	2022	iProm-phage	Link	–	One-hot	CNN	5-fold CV	Promoter & Phage Classification	Bacteriophages	phiSITE	99 bp
Deep Learning	2022	PromoterLCNN	Not Available	Link	One-hot Encoding	2 Stage CNN	5-fold CV	σ^{70}, σ^{24}, σ^{32}, σ^{38}, σ^{28}, σ^{54}	E. Coli	RegulonDB	81 bp
Deep Learning	2023	iProm-Sigma54	Link	Link	One-hot Encoding	1D CNN	5-fold CV	Promoter or Not	σ^{54}	Pro54DB	81 bp
Deep Learning	2023	iPro-TCN	Not Available	–	Word2Vec	TCN	–	Promoter and Strength	E. Coli	RegulonDB	∼45 bp
Deep Learning	2024	iProL	Not Available	Link	k-mer at k=2	CNN + BiLSTM	5-fold CV	Promoter or Not	E. Coli k-12	RegulonDB	81 bp
Deep Learning	2024	msBERT-Promoter	Not Available	Link	k-mer at k=3,4,5,6	Transformer	–	Promoter and Strength	E. Coli	RegulonDB	81 bp
Deep Learning	2024	iPSI(2L)-EDL	Link	–	One-Hot + NCPD	CNN + LSTM	–	Promoter and Strength	E. Coli	RegulonDB	81 bp
Machine Learning	2024	CDBProm	Link	Link	DDS	XGBoost	–	Promoter or Not	Multiple Genome	–	–
Deep Learning	2024	iPro2L-DG	Not Available	Link	C2 & NCP	CNN + Attention	5-fold CV	Promoter and Strength	E. Coli k-12	RegulonDB	81 bp
Deep Learning	2025	ProPr54	Link	Link	One-Hot	CNN + BiLSTM + Attention	–	σ^{54}	Multiple Genome	–	50 bp
Deep Learning	2025	PromoterAtlas	Not Available	Link	Raw Sequence used	Transformer	–	Promoter or Not	Multiple Genome	–	–
Deep Learning	2025	pPromoter-FCGR	Not Available	–	Frequency Chaos Game Representation	CNN-based models	–	Promoter or Not	E. Coli	RegulonDB	300 bp

Table 1: Overview of bacterial promoter prediction frameworks and tools

The story behind the prokaryotic promoter-prediction programs is an interesting one. Beginning with simple, machine-learning models, the pendulum has steadily swung toward today's state-of-the-art deep-learning techniques. The table catalogues more than 30 diverse frameworks developed between 2016 and 2025, most created in order to annotate promoter regions — important regulatory elements within bacterial genomes. All tools each have their own style of sequence encoding, algorithms and testing approaches in an overall common aim to refine the accuracy of promoter detection.

When you examine the tools over time, a pattern is revealed. In the early days, many models relied on handcrafting features but now it is possible to learn from data and automatically discover patterns through deep representation learning. The focus has also broadened from tools designed to work with just a single species to those that operate across many genomes. And more and more, model architectures are not shallow classifiers, but instead complex patterns tend to be learned over sequences in clever ways. Taken together, this progress is propelling promoter prediction into a new age; one that can provide higher accuracy, wider scope and clearer biological understanding.

Eukaryotic Promoters

Framework	Year	Tool	Web-server	Github	Embedding	Algorithm	Evaluation strategy	Promoter Type	Species	Database	Sequence Length
Machine Learning	2016	SD-MSAEs	Not Available	–	n-mer	Ensemble of SVM Classifier	10-fold CV	Promoter or Not	Homo Sapiens	DBTSS	81 bp
Machine Learning	2017	TSSPlant	Not Available	Link	–	ANN	–	Both TATA & TATA-less Promoter	Plant	ppdb & PlantProm	600 bp
Machine Learning	2017	probabilistic integrative algorithms-driven model	Not Available	–	CG-skew, TATA motif frequency, frequency of nucleotide	Binary Classifier	–	Both TATA & TATA-less Promoter	Rice Plant (Oryza sativa)	RNA-Seq Dataset	300 bp
Machine Learning	2018	PromPredict	Not Available	–	DNA duplex free energy	Sliding window segments	–	Both TATA & TATA-less Promoter	48 eukaryotic genomes	–	1001 bp
Deep Learning	2019	EPI Prediction	Not Available	Link	One-hot	CNN	–	Interaction of enhancers and promoters	Human	Hi-C EPI datasets	3000 bp for enhancer & 2000 bp for Promoter
Statistical-based Method	2020	Statistical alignment method	Not Available	–	Encoding via construction of PWMs	PWM Classification	–	Both TATA & TATA-less	Arabidopsis thaliana	EPD	600 bp
Statistical-based Method	2021	MAHDS	Not Available	–	–	PWMs & Profile matrix	–	Promoter or Not	Plants	Independent data	600 bp
Deep Learning	2021	Cr-Prom	Link	Link	One-hot	CNN	5-fold CV	Promoter or Not	Rice	PlantProm DB	251 bp
Deep Learning	2021	iPTT(2L)-CNN	Link	–	One-hot	CNN	5-fold CV	TATA & TATA-less promoter	Plants	Independent data	251 bp
Deep Learning	2022	iProm-Zea	Link	–	One-hot	CNN	5-fold CV	TATA & TATA-less Promoter	Plant	Independent Data	251 bp
Deep Learning	2022	DeeProPre	Not Available	Link	Supervised Embedding	CNN + BiLSTM + Attention	5-fold CV	Promoter or Not	Human & Mouse	EPD	251 bp
Deep Learning	2022	iPromoter-Seqvec	Not Available	Link	Sequence-embedded features	BiLSTM	–	TATA & TATA-less Promoter	Human & Mouse	EPD	300 bp
Machine Learning	2023	PromGER	Not Available	Link	Graph Embedding	Tree-base ensemble Classification	–	TATA & TATA-less Promoter	H. Sapiens, R. norvegicus, Z. Mays	EPD, DBTSS, EID	251 bp, 300 bp, 1001 bp
Deep Learning	2025	hybrid model combining DNA2Vec embeddings + DNN	Not Available	–	DNA2Vec with UNK	DNN	–	Promoter or Not	Human	EPDnew	600 bp
Deep Learning	2025	OntoGene	Not Available	Link	Sequence & Knowledge graph	BERT	5-fold CV	Promoter or Not	Human	Deciptor Dataset & Gene Ontology	3000 bp

Table 2: Overview of eukaryotic promoter prediction frameworks and tools

Eukaryotic & Prokaryotic Promoters

Framework	Year	Tool	Webserver	Github	Embedding	Algorithm	Evaluation Strategy	Promoter Type	Species	Database	Sequence Length
Deep Learning	2017	CNNProm	Link	Link	One-hot	CNN	–	σ^{70}, TATA & TATA-less promoter	E. Coli, Human, Mouse, Arabidopsis	RegulonDB, EPD, DBTBS	For bacteria: 81 bp, For other: 252 bp
Machine Learning	2019	iproEP	Link	–	PseKNC & PCSF	SVM	10-fold CV	Promoter or Not	E. Coli, B. Subtilis, H. Sapiens, C. Elegans, D. melanogaster	EPD, GenBank, Independent datasets	81 bp for prokaryotic & 300 bp for eukaryotic
Statistical-based method	2021	IPMD	Not Available	–	PCSF & Diversity Increment	Modified Mahalanobis Discriminant classifier	–	Promoter or Not	E. Coli, B. Subtilis, H. Sapiens, C. Elegans, D. melanogaster	EPD, GenBank, RegulonDB, DBTBS	81 bp for prokaryotic & 300 bp for eukaryotic
Deep Learning	2022	CapsProm	Not Available	Link	Nucleotide frequency + One-hot	Capsule Network	5-fold CV	Promoter or Not, TATA & TATA-less	E. Coli, B. Subtilis, H. Sapiens, A. thaliana, M. musculus	EPDnew, GenBank, RegulonDB, DBTBS	81 bp for prokaryotic & 251 bp for eukaryotic
Deep Learning	2022	iPro-WAEL	Not Available	Link	PseKNC, PseDNC, K-mer, CKSNAP, Word2Vec	RF, CNN	–	Promoter or Not, TATA & TATA-less	E. Coli, B. Subtilis, Human, A. thaliana, Mouse, etc.	13 different datasets	81 bp for prokaryotic & 3000 bp for eukaryotic
Deep Learning	2025	HybProm	Not Available	Link	DNA2Vec	CNN, BiLSTM, Attention	–	Promoter or Not, TATA & TATA-less	E. Coli, Human, Mouse, Plant	EPD, DBTSS, RegulonDB	81 bp for prokaryotic & 251 bp for eukaryotic
Deep Learning	2025	iPro-CSAF	Not Available	Link	Raw sequence	CSNN + Spiking Attention	–	Promoter or Not, TATA & TATA-less	E. Coli, B. Subtilis, Human, A. thaliana, Mouse, etc.	Multiple Datasets	81 bp for prokaryotic & 251 bp for eukaryotic

Table 3: Overview of combined prokaryotic and eukaryotic promoter prediction frameworks and tools

Archaea Promoters

Framework	Year	Tool	Webserver	Github	Embedding	Algorithm	Evaluation Strategy	Promoter Type	Species	Database	Sequence Length
Machine Learning	2022	Classification Models for Archaeal	Not Available	Link	DNA duplex stability	SVM	10-fold CV	Promoter or Not	H. Volcanii, S. solfataricus, T. kodakarensis	RSAT prokaryotics database	101 nt
Machine Learning	2023	Classification Models for Archaeal	Not Available	Link	DNA duplex stability	SVM	10-fold CV	Promoter or Not	H. Volcanii, S. solfataricus, T. kodakarensis, T. Pendens, A. Boonei	Transcriptome-mapped archaeal data	101 nt
Deep Learning	2025	iProm-Archaea	Link	Link	One-hot, NCP, DDS, PseEIIP, PseDNC	CNN	5-fold CV	Promoter or Not	H. Volcanii, S. solfataricus, T. kodakarensis	Experimentally Verified	101 nt

Table 4: Overview of archaeal promoter prediction frameworks and tools